



Ahmedabad
University

GROUP 17

Probabilistic Adaptive Traffic Signal Control in Intelligent Transport Systems (ITS)

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Project Context and Motivation

Exact Objective:

To develop a simulation pipeline that evaluates deterministic traffic controllers and quantifies their inefficiency under stochastic traffic conditions.

What is being decided?

The timing of signal phase transitions (Green and Red durations).

What is being estimated?

Real-time queue length and waiting time across different times of the day.

Scope:

4-way intersection, 8 directional movements, comparing a Naive Fixed-Timer and a Time of Day (ToD) baseline.

Project System Overview:

Demand Generation:

Python scripts generating Non-Homogeneous Poisson arrivals.

Simulation Engine:

SUMO handles vehicle physics and physical intersection rules.

Interface:

TraCI API extracts real-time sensor data from the simulation.

Baseline Controllers:

Tested both a Naive Fixed-Timer and a Time-of-Day (ToD) schedule.

Data Flow:

Time-varying Arrival Rates flow to SUMO, TraCI extracts Sensor Data, exporting to Queue Logs for Python Statistical Analysis.

Sources of Uncertainty in the Project

Measurement Noise:

TraCI sensor detection limits, such as occlusion when vehicles are too close together.

Environmental Variability:

Traffic volume changes throughout the day. We modeled 5 distinct phases:

- Off-Peak
- Morning Peak
- Mid-Day
- Evening Peak
- Night

Modeling Uncertainty:

Sudden clumping of vehicles (high variance) occurs randomly even within pre-defined peak hours, challenging fixed timers.

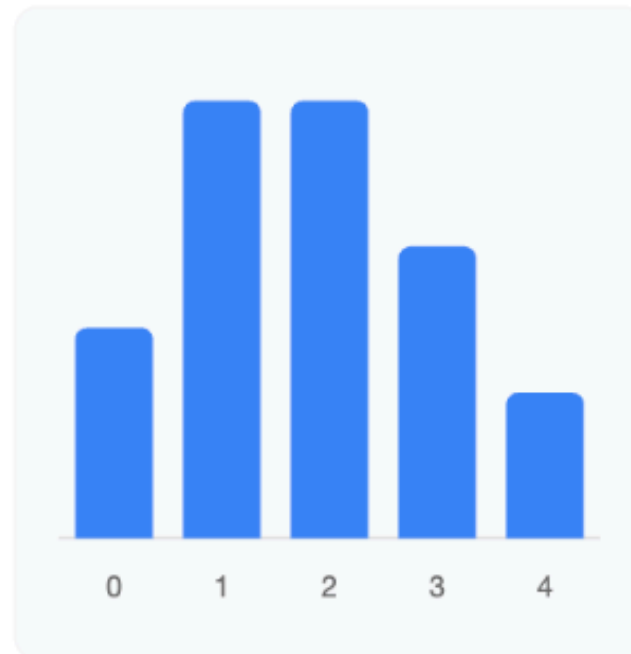
Key Random Variables and Their Roles

STOCHASTIC INPUT

Arrivals N_k

$$N_k \sim \text{Poisson}(\lambda \cdot \Delta t)$$

Number of vehicles arriving in interval Δt . Follows a Poisson distribution characterized by mean arrival rate λ .

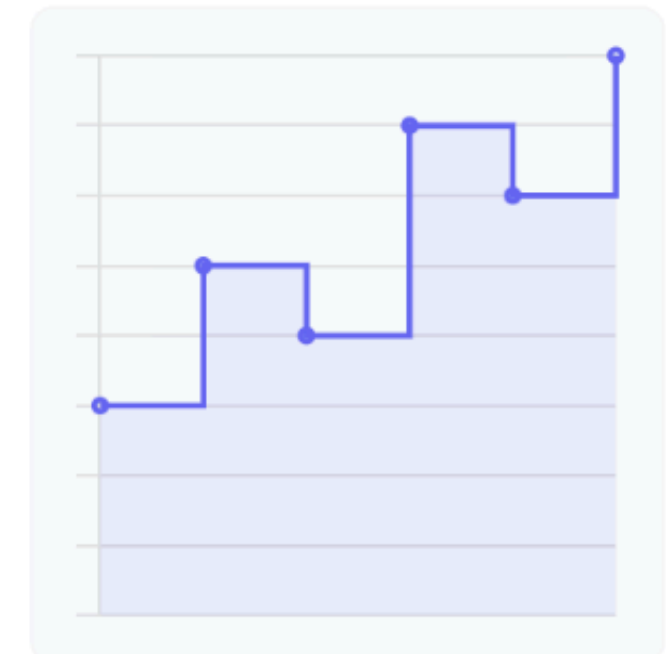


SYSTEM STATE

Queue Q_k

$$Q_{k+1} = \max(0, Q_k + N_k - q_{out})$$

Modeled as a **Reflected Random Walk**. The queue grows with arrivals and shrinks with departures, never dropping below zero.



BINARY CLASSIFIER

Congestion S_k

$$S_k = \mathbb{1}(Q_k \geq T)$$

A simplified binary state (0 or 1). Active when queue length exceeds threshold T . Helps reduce state space complexity.

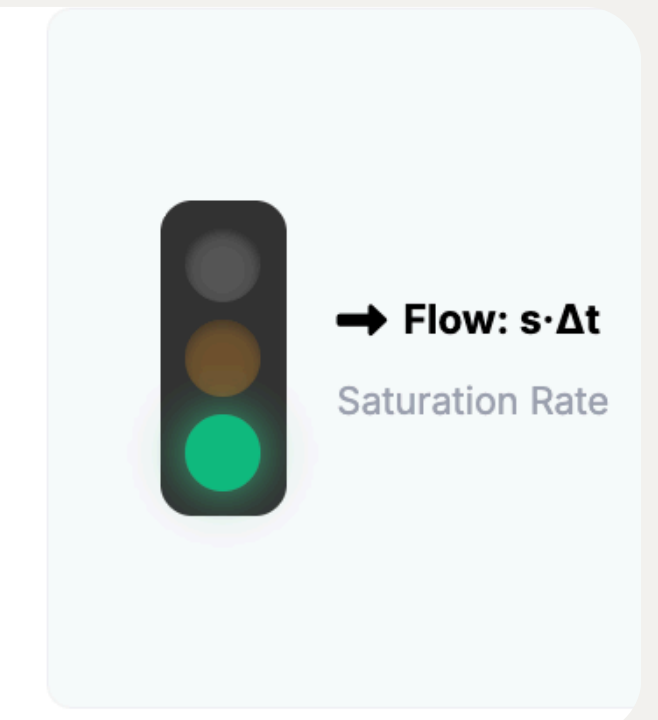


CONTROLLED OUTPUT

Departures q_{out}

$$q_{out} = \min(s \cdot \Delta t, Q_k) \cdot \mathbb{1}(\text{Green})$$

Depends on control $u(k)$. Zero if Red. If Green, limited by saturation flow s or available queue.



Probabilistic Models and Assumptions

Distribution:

Non-Homogeneous Poisson Process. The arrival rate changes based on the time of day to simulate realistic rush hours.

Independence Assumption:

Vehicle arrivals in the North-South lane do not influence East-West arrivals.

Controller Assumption:

Both the Fixed and ToD baseline controllers are blind to the actual queue; they operate purely on the clock.

Justification:

Poisson modeling accurately captures the realistic clumping of traffic, highlighting exactly why fixed timers fail.

Probabilistic Reasoning and Inference Logic

Propagation:

Uncertainty in arrivals feeds into the queue evolution equation.
Surges overwhelm the fixed departure rate.

Reasoning Pathway Step 1:

Observe queue growth during peak hours.

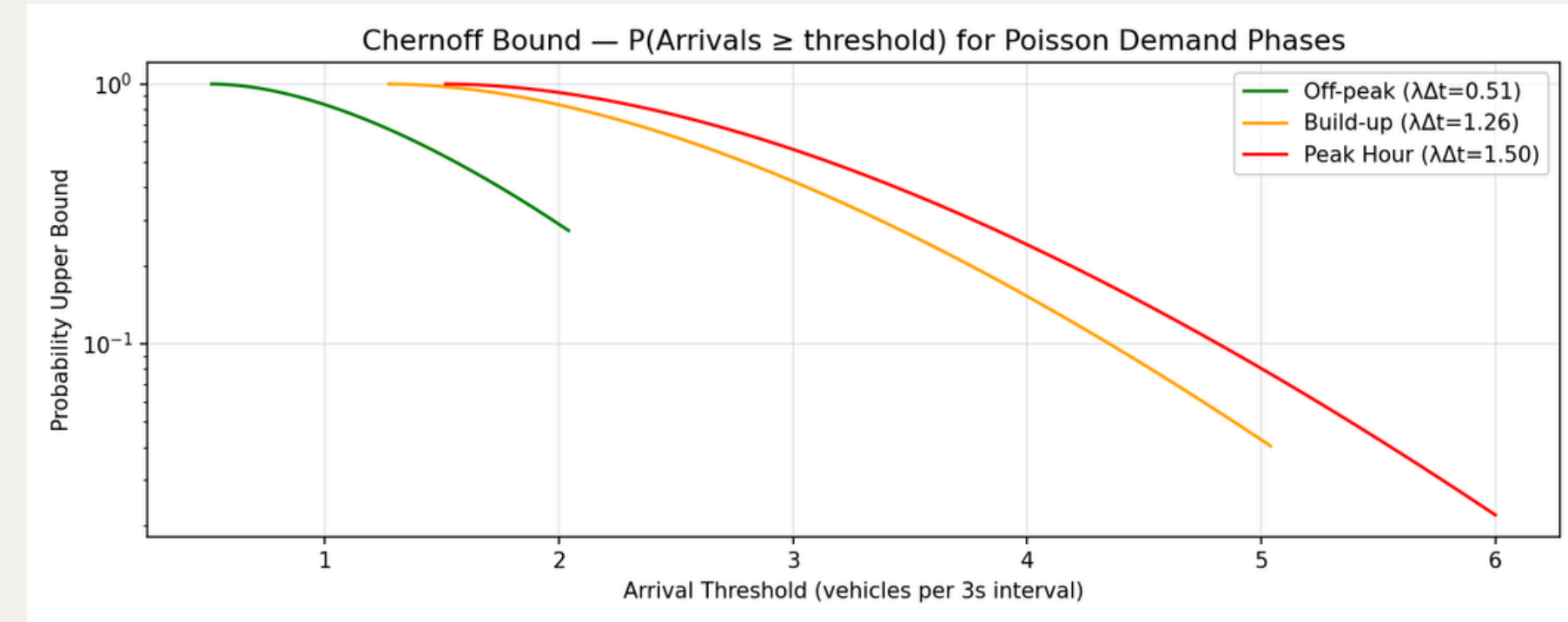
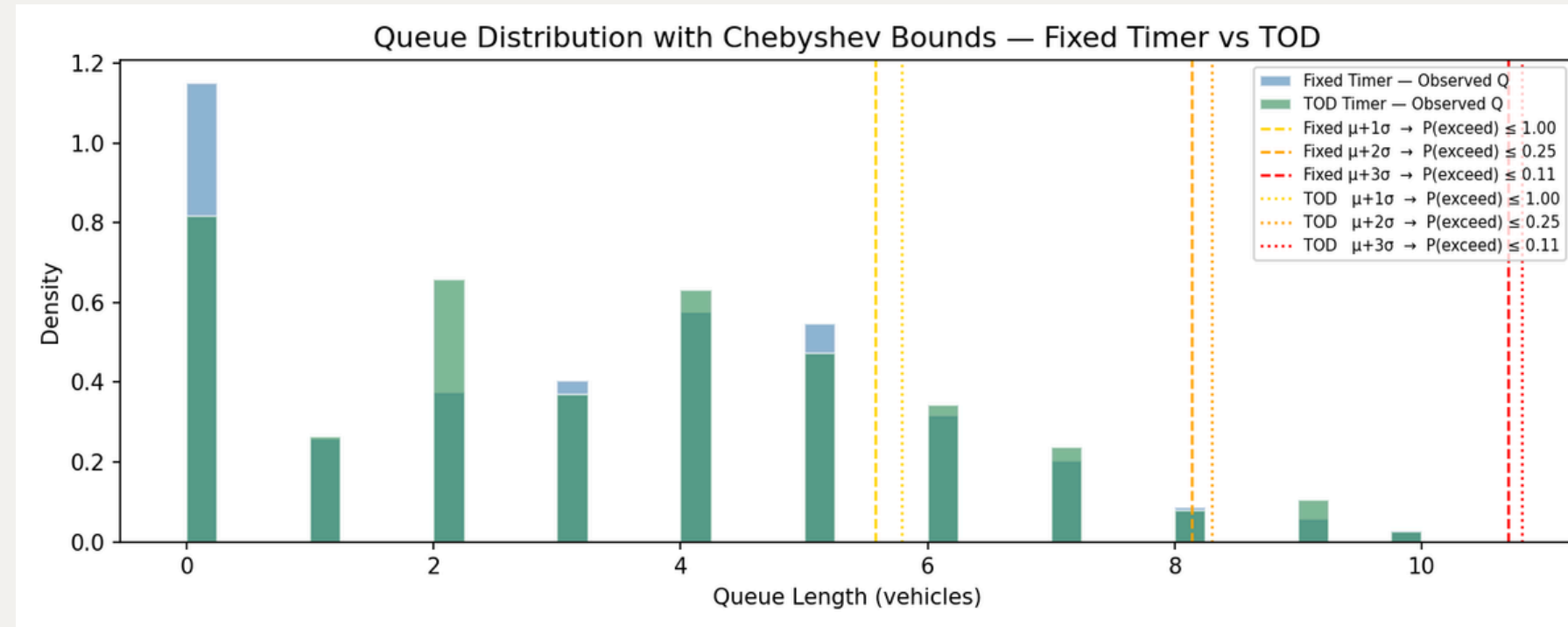
Reasoning Pathway Step 2:

Apply the Chebyshev Inequality to identify extreme congestion events.

Reasoning Pathway Step 3:

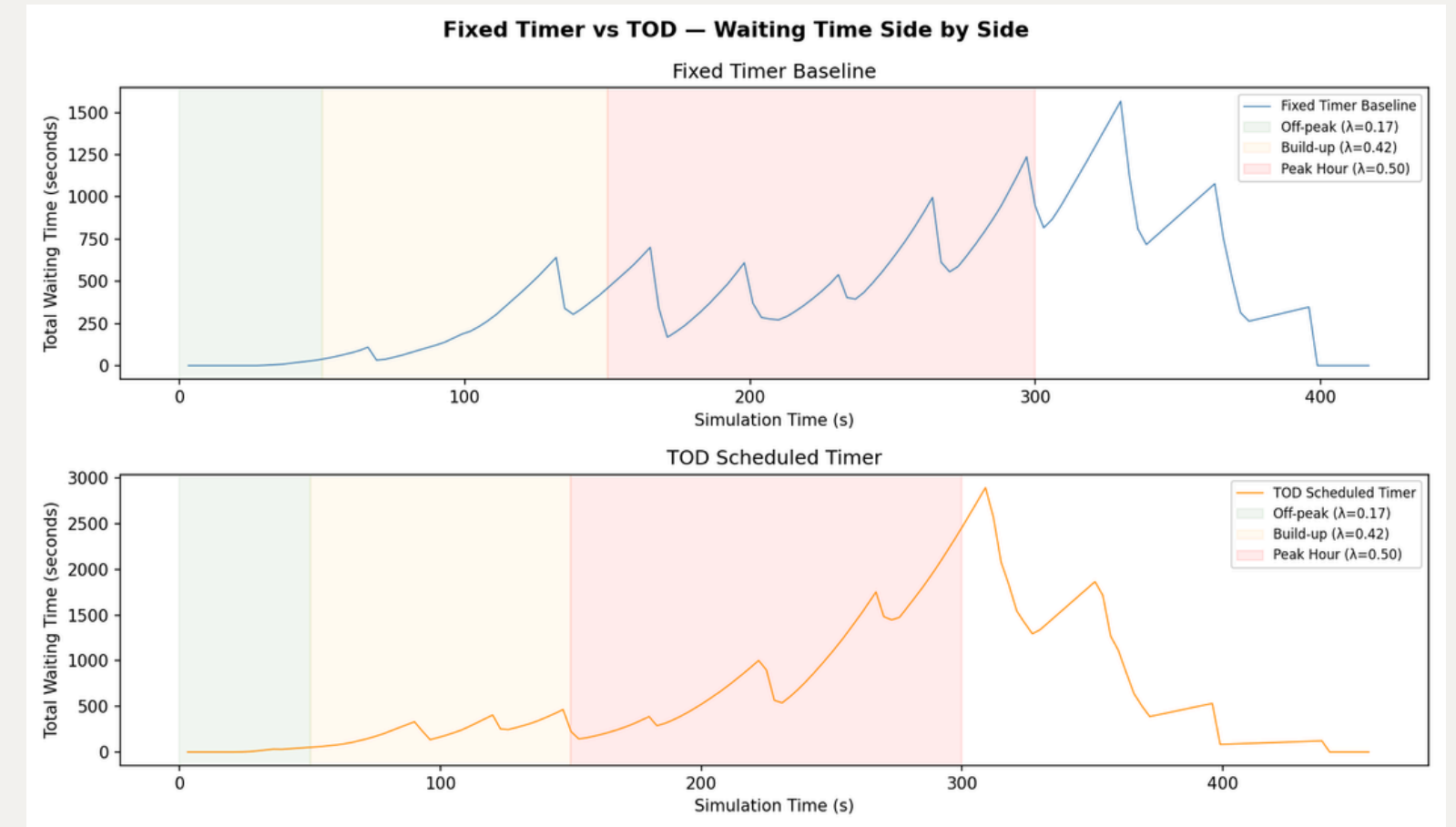
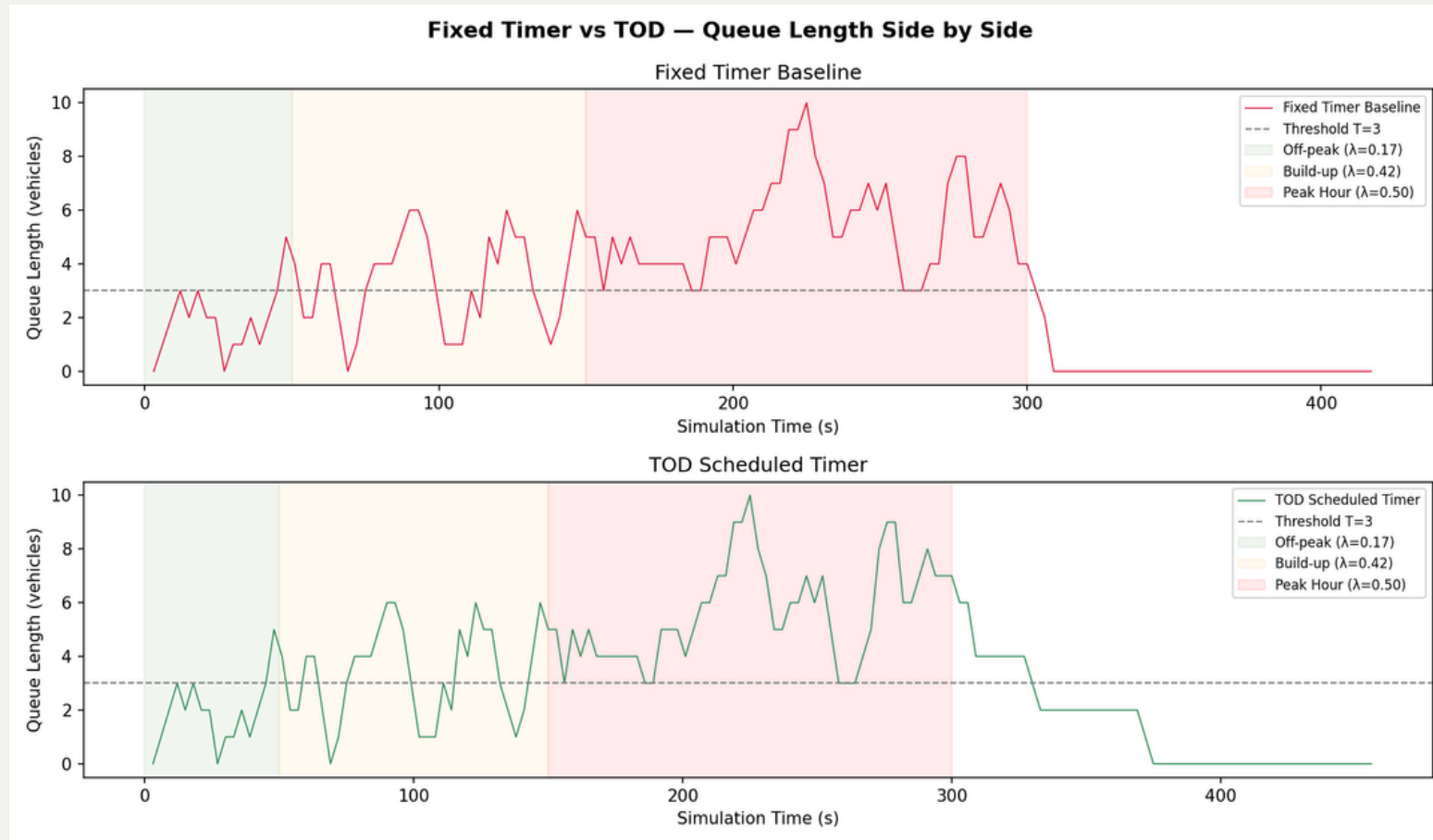
Conclude that average-based timing cannot handle the tail-end risks of traffic surges.

Probabilistic Bounds Validation — Chebyshev & Chernoff



- Chebyshev bounds extreme queue events ($P(Q \geq \mu+3\sigma) \leq 0.11$).
- Chernoff tightens tail probability for Poisson arrivals across peak phases.

Simulation Results — Fixed Timer vs TOD Controller



- TOD reduces mean queue by 25% and waiting time by 40%, but variance remains high.
- Deterministic controllers fail during stochastic traffic surges.

Operationalizing Reasoning in Simulation

Realization:

We ran the simulation against two baselines: a strict Fixed timer and a Time-of-Day (ToD) timer.

Evaluation Metrics:

We tracked Total Queue Length and Waiting Time step-by-step.

Observation:

During the Evening Peak, both deterministic systems experienced massive spikes because traffic variance ignores average schedules.

Operationalizing Reasoning in Simulation

Weakness:

Even the improved Time-of-Day (ToD) controller is blind. It changes based on the clock, wasting Green time on empty lanes.

Unverified Assumptions:

We assume saturation flow is constant; in reality, driver acceleration and vehicle types vary.

Ambiguity:

Setting a static threshold for extreme congestion does not perfectly capture the fluid nature of varying traffic density.

Operationalizing Reasoning in Simulation

Evolution for Milestone 4:

Transitioning from Deterministic Baselines to a fully Adaptive Markov Decision Process.

Conceptual Tasks:

Coding the Bellman Equation to optimize signal phases dynamically based on real-time TraCI queue data.

Role Coordination:

GitHub Manager ensures version control for the dynamic SUMO files; Reviewer validates the probabilistic logic in the adaptive script.

Summary of Group-Level Understanding

The Big Picture:

Our simulation proved mathematically and visually that Averages are a Trap.

Synthesis:

Adjusting timers for time-of-day is not enough. The inherent variance in traffic causes massive delays under any deterministic rule.

Next Steps:

We have established a robust reference point and are ready to implement a real-time probabilistic controller to eliminate these spikes.